SMART WALKING STICK FOR VISUALLY IMPAIRED NAVIGATION WITH AI INTEGRATION

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Project Proposal Report

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DECLARATION

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We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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....30/Aug/2025....

Signature of the Supervisor Date

(Dr. Junius Anjana)

ABSTRACT

Visual impairment poses significant safety risks for approximately 2.2 billion people worldwide, particularly in emergency situations where rapid response is critical. Traditional assistive devices like white canes lack the ability to detect vocal distress signals, leaving users vulnerable during falls, assaults, or health crises. This component of the Smart Walking Stick project addresses the need for automated emergency detection through voice analysis to enhance user safety and independence.

The proposed emergency voice detection system integrates a microphone module with TensorFlow Lite machine learning models to identify distress phrases (e.g., "help me") or shouting in real-time. The system captures audio input, processes it using a trained TinyML model for stress detection, and triggers immediate SMS alerts to caregivers via Twilio API. The component focuses on detecting key distress indicators including elevated pitch, volume spikes, and specific keywords.

The research employs audio processing techniques, edge computing, and human-computer interaction principles to develop a lightweight, power-efficient solution that operates in real-time. The system ensures minimal latency while maintaining high accuracy in voice stress detection, making it suitable for integration into a portable assistive device. Expected outcomes include improved emergency response times, enhanced safety monitoring, and increased confidence for visually impaired users in various environments.

Keywords: Voice Stress Detection, Audio Processing, TinyML, Microphone Integration, Assistive Technology, Visual Impairment

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
CNN	Convolutional Neural Network
ETA	Electronic Travel Aid
FPS	Frames Per Second
IoT	Internet of Things
MTBF	Mean Time Between Failures
PCB	Printed Circuit Board
Tiny ML	Tiny Machine Learning
FR	Functional Requirements
NFR	Non-Functional Requirement

1. INTRODUCTION

Visual impairment significantly impacts personal safety and independence, affecting over 2.2 billion individuals globally according to the World Health Organization [1]. Emergency situations, such as falls, medical issues, or external threats, can be particularly dangerous for visually impaired people who may struggle to call for help effectively. Traditional mobility aids like white canes provide no mechanism for detecting vocal distress or automating emergency alerts [2].

The integration of IoT sensors, audio processing, and machine learning offers promising solutions for intelligent emergency detection. Advances in edge computing and TinyML enable real-time audio analysis on resource-constrained devices, allowing for immediate response without relying on external networks [3].

Emergency voice detection is crucial for enhancing user safety by identifying distress signals like shouting or phrases such as "help me." This capability can trigger automated alerts to caregivers, reducing response times in critical situations [4]. The microphone module, combined with Bluetooth transmission and API integration, provides a robust platform for implementing such systems [5].

This research component focuses on developing an emergency voice detection system using a microphone and TensorFlow Lite models to detect distress and send SMS alerts via Twilio API. The system advances assistive technology by combining audio AI, embedded systems, and emergency response protocols to create an effective safety feature for the Smart Walking Stick.

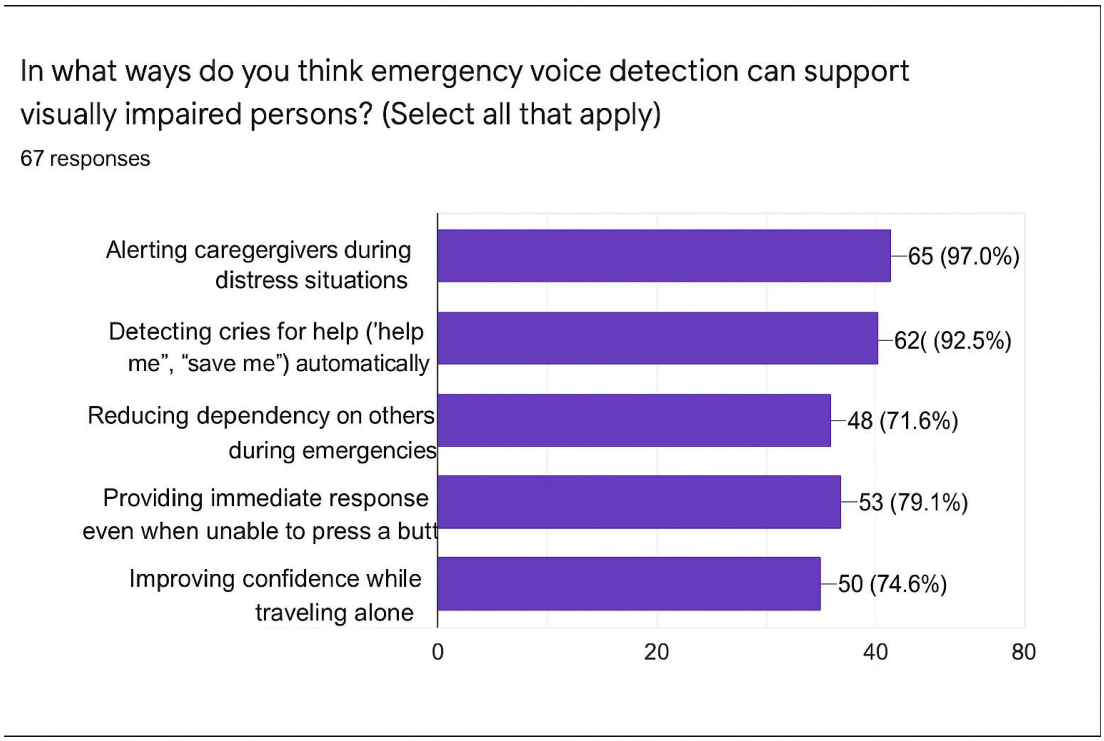

Figure 1.1 what ways do you think emergency voice detection can support visually impaired persons

1.1 Background & Literature Survey

1.1.1 Assistive technologies for visual impairment

Assistive technologies have progressed from basic mechanical aids to sophisticated electronic systems. Early devices focused on mobility, but modern solutions incorporate sensors for environmental awareness [6]. However, emergency detection remains underdeveloped in most aids [7].

1.1.2 Audio processing in assistive technology

Audio processing has been applied in various assistive contexts, such as speech recognition for navigation. Voice stress analysis uses features like pitch and amplitude to detect emotions [8]. CNNs have proven effective for audio classification tasks [9]. Lightweight models like MobileNet enable deployment on embedded devices [10].

1.1.3 Edge computing and TinyML

TinyML facilitates AI on microcontrollers, enabling real-time audio processing without cloud dependency [11]. TensorFlow Lite supports audio model optimization through quantization [12].

1.1.4 Microphone integration in embedded systems

Microphone modules have been used in IoT devices for voice commands. Integration with Arduino and Bluetooth allows seamless data transmission [13]. Research shows successful audio distress detection in wearables [14].

1.1.5 Emergency alert systems

Systems like Twilio API enable SMS alerts in health monitoring devices. Voice-based emergency detection has been explored in smart homes but less in portable aids [15]. Integration with mobile apps enhances response capabilities [16].

1.1.6 Human computer interaction in assistive technology

Effective HCI for visually impaired users relies on haptic and audio feedback. Automated alerts must be reliable to avoid false positives [17]. Vibrotactile patterns can complement voice detection for confirmation [18].

1.2 Research gap

Current emergency detection systems for visually impaired individuals have several limitations:

1.2.1 Limited real-time audio processing

Many systems rely on cloud processing, causing delays in emergencies. Edge-based solutions are needed for instant detection.

1.2.2 Lack of integrated portable solutions

Emergency features are often standalone apps, not embedded in mobility aids.

1.2.3 Ignored vocal tension signals

Emphasis on body sensors overlooks sound markers like pitch changes. Most smart canes focus on physical sensors for fall or obstacle detection, neglecting vocal cues such as shouting or distress phrases that could indicate emergencies like assaults or medical issues. This omission leaves a critical vulnerability in safety protocols, as voice-based indicators are often the first and most natural response in urgent situations.

1.2.4 Adjustment deficiencies

Restricted management of dialects, interference, or tongues. Existing voice-enabled systems rarely account for environmental noise, accents, or multilingual support, leading to high false negative rates in diverse real-world settings. This lack of adaptability reduces reliability for users in noisy urban areas or non-English speaking regions, underscoring the need for robust, context-aware audio processing.

1.2.5 Inferior alerts

Notifications miss strong guardian linkage. While some devices include emergency alerts, they often lack integration with reliable communication APIs for automated caregiver notifications, resulting in delayed responses. This gap in seamless alert mechanisms fails to provide the rapid intervention necessary for user safety in critical moments.

1.2.6 Expense problems

Elevated prices limit reach. Advanced smart canes with any form of emergency features are often prohibitively expensive due to complex hardware, restricting accessibility for low-income visually impaired populations. This economic barrier prevents widespread adoption and highlights the demand for cost-effective, scalable solutions using affordable components.

Table 1.1 comparison of existing smart stick systems with the proposed system

Feature / System	We WALK Smart Cane (Research 1)	Smart Cane IIT Delhi (Research 2)	IoT-Based Smart Stick (Research 3)	A IoT Smart Stick (Research 4)	Proposed Research
Real-time Landmark Recognition	✕	✕	✕	✕	✓
Embedded Processing	✕	✓	✓	✕	✓
Environmental Landmark Detection	✕	✕	✕	✕	✓
Multimodal Feedback (audio + haptic)	✓ (audio only)	✓ (vibration only)	✓ (audio only)	✓ (audio only)	✓ (audio + haptic)
Portable / Integrated	✓	✓	✓	✓	✓
Affordable / Accessible	✕	✓	✓	✕	✓

1.3 Research problem

Visually impaired individuals encounter heightened vulnerabilities in emergency situations due to the limitations of traditional assistive devices in detecting and responding to vocal distress signals. Conventional white canes and similar tools provide no mechanism for audio-based emergency detection, leading to potential delays in assistance that can exacerbate risks during falls, assaults, or medical crises. The absence of integrated voice stress detection in current technologies creates several critical issues:

1.3.1 Emergency response delays

Without automated detection of distress calls, visually impaired users often experience prolonged wait times for help, leading to increased risk of injury, difficulty in summoning immediate assistance, greater dependence on bystanders, and reduced overall safety in urgent scenarios.

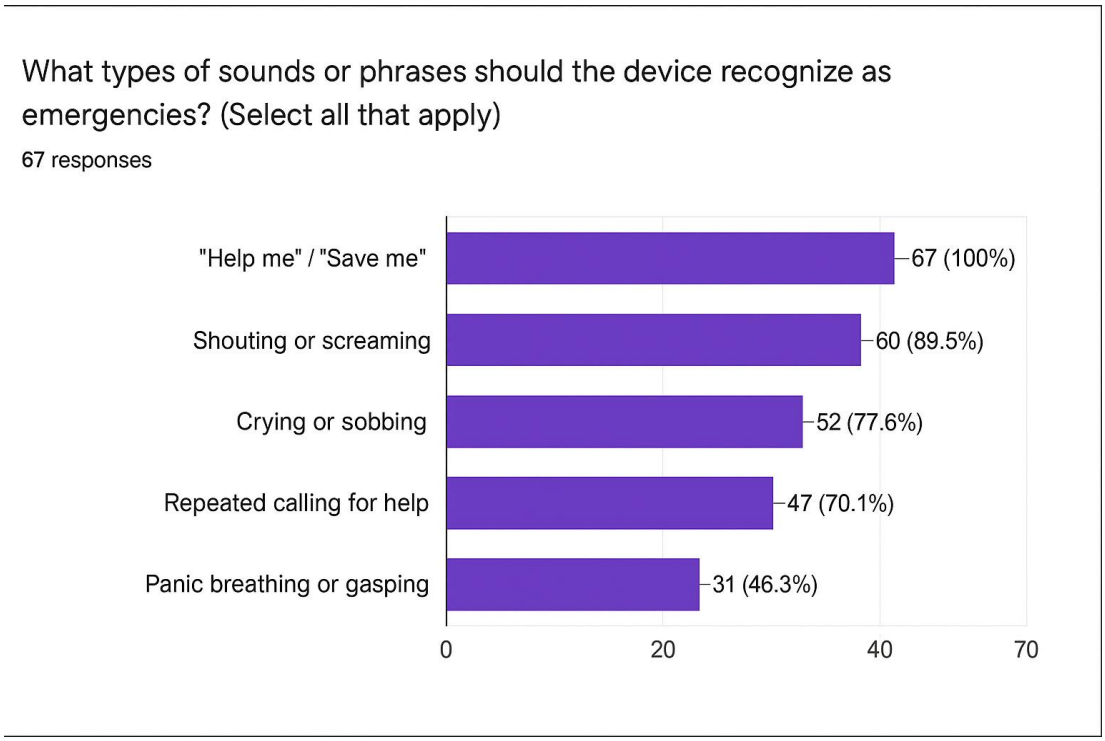

Figure 1.2 What types of sounds or phrases should the device recognize as emergencies

1.3.2 Lack of vocal cue detection

Current mobility aids fail to recognize audio indicators of distress such as shouting or specific phrases, resulting in missed opportunities for early intervention, limited awareness of personal emergencies, inability to detect subtle stress in voice patterns, and incomplete protection against unforeseen threats.

1.3.3 Limited situational awareness

Existing technologies provide minimal feedback on emergency contexts through audio, causing lack of proactive alert systems, inability to integrate with caregiver networks, reduced confidence in independent mobility, and inadequate handling of

Figure 1.2 How landmark detection supports visually impaired users in safe and efficient navigation.

complex crisis situations.

1.3.4 Technology limitations

Present solutions face challenges including reliance on external networks for processing, high latency in alert generation, requirement for sophisticated hardware, and poor performance in noisy environments.

1.3.5 User experience challenges

Systems often lack intuitive interfaces for emergency features, with complicated activation methods, overwhelming notifications, limited customization for different needs, and insufficient support for diverse languages or accents.

The research problem can be formulated as follows:

"How can we develop an intelligent, embedded emergency voice detection system that provides real-time identification of vocal distress to enhance safety and independence for visually impaired individuals, while addressing the limitations of current assistive technologies in terms of accessibility, reliability, and cost?"

This problem encompasses the technical challenges of implementing audio processing algorithms on resource-constrained devices, the design challenges of creating effective alert mechanisms, and the practical challenges of building an affordable and user-friendly emergency feature.

2. OBJECTIVES

2.1 Main objective

To build a microphone-oriented crisis identification system with TensorFlow Lite that recognizes vocal tension and sends Twilio texts for heightened protection in visually impaired travel.

2.2 Specific objectives

2.2.1 Machine learning model crafting

- Craft TensorFlow Lite model for stress classification achieving 88% accuracy
- Apply quantization for device optimization
- Assemble audio datasets with distress samples
- Assess in varied acoustic environments

2.2.2 Sensor embedding

- Embed microphone for ongoing audio sampling
- Tune for urban noise rejection
- Build feature extraction flows

2.2.3 Connectivity setup

- Enable Bluetooth data relay to apps
- Guarantee minimal delay transfers
- Incorporate fault tolerance

2.2.4 Notification framework

- Link Twilio for customizable SMS
- Configure location-inclusive messages
- Verify with test recipients

2.2.5 Assembly and evaluation

- Merge with cane architecture
- Perform field trials
- Measure detection speed and precision

2.2.6 Efficiency tuning

- Reduce energy draw
- Lower false detections
- Bolster environmental adaptability

3. METHODOLOGY

3.1 Overall system description

This unit enhances the Smart Walking Stick with sound tension oversight: microphone entries go into TinyML for grouping, activating program-oriented alerts. Imagined chart shows path from recording to alert.

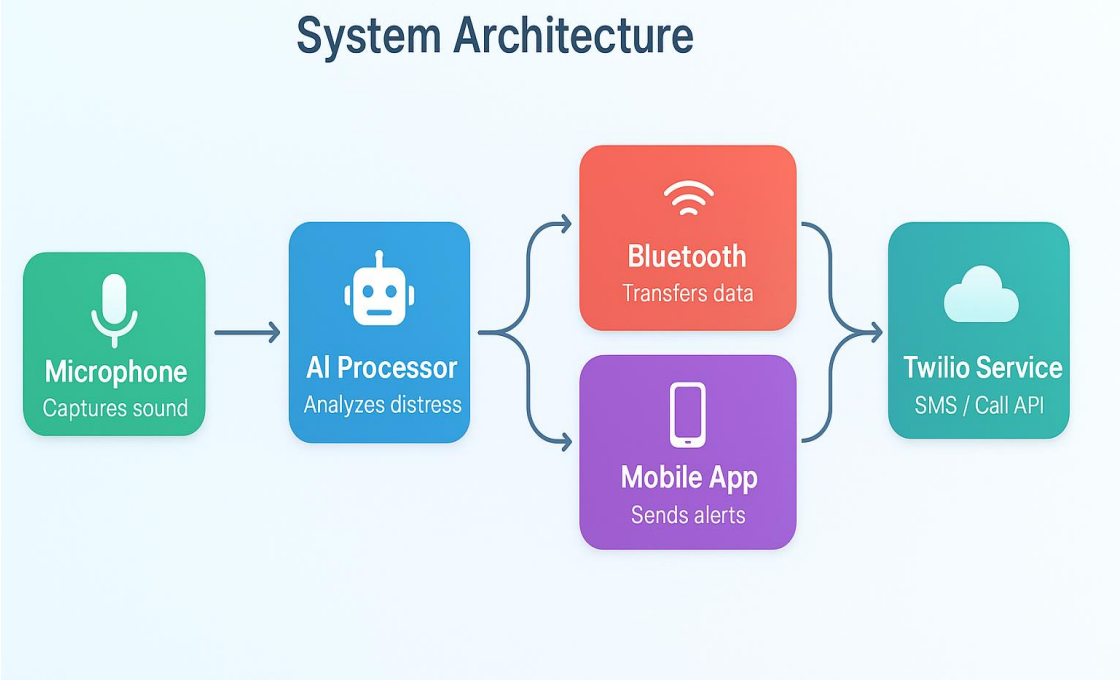

Figure 3.1 System Architectural Diagram

3.2 Development methodology

Using agile cycles for quick building and response-based improvements.

3.2.1 Ghantt Chart

The Gantt chart for the "Smart Walking Stick for Visually Impaired" project outlines a structured timeline from August 2025 to June 2026. It details key phases, including planning, design, and model development, with specific tasks such as project initiation, literature review, hardware/software architecture design, and mobile app development. Each task is color-coded and scheduled to ensure progress, quick prototyping, and continuous refinement based on user feedback and testing results.

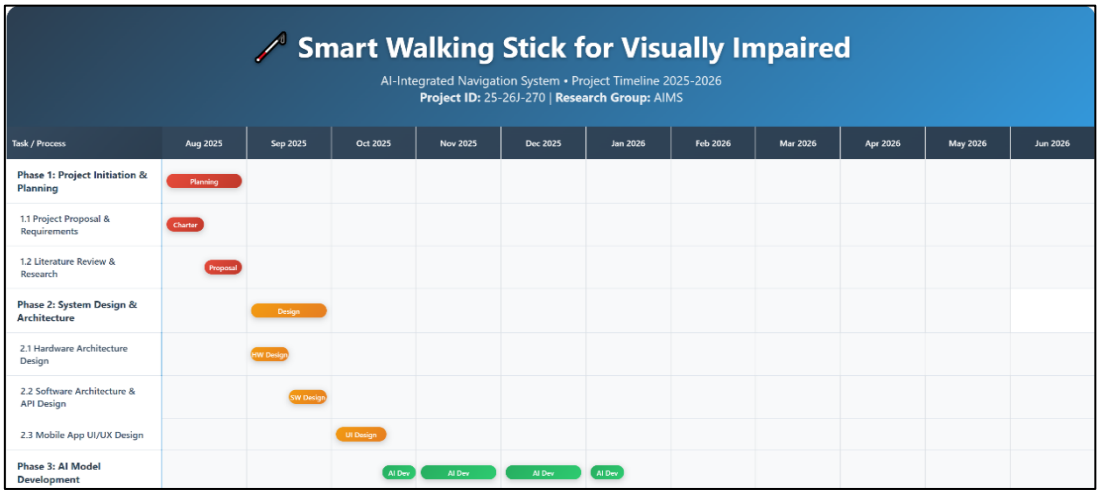

Figure 3.2 Gantt Chart for the entire research project

Full hosted Ghantt chart for the whole year - https://ghant-chart-25-26j-270.netlify.app/

3.2.2 Work Breakdown Structure

Phase 1: Planning (Weeks 1-5)

- Read studies and collect info
- Buy needed parts

Phase 2: Build AI model (Weeks 6-13)

- Prepare sound data and train the model
- Make it better through tests

Phase 3: Hardware Implementation (Weeks 13-20)

- Code the microphone sensor
- Test connections like Bluetooth

Phase 4: Combine everything (Weeks 22-29)

- Link alerts to the app
- Check the whole system works

Phase 5: Test and fix (Weeks 30-36)

- Try it with real users to see how it performs in daily life.
- Write up the results and lessons learned for the report.

3.3 Technical implementation approach

3.3.1 ML model creation

- Collect tension sound (e.g., from ESC-50, own records)
- Use LSTM-CNN mix
- Enhance with interference layers

3.3.2 Microphone configuration

- Arrange for 16kHz recording
- Use screens for disruption

3.3.3 Communication system

- Bluetooth LE for effective transfer

3.3.4 Feedback system design

- Twilio SDK for location-marked texts

3.4 Testing and validation methodology

Controlled Indoor Testing

- Purpose: To establish a baseline performance in a controlled environment with minimal external interference.
- Procedure: Conduct tests in a quiet indoor setting (e.g., a soundproof room) using pre-recorded distress sounds (e.g., shouts of "help," raised voices) and simulated background noise (e.g., white noise, faint conversations).
- Equipment: Use a high-quality microphone and speaker system to play audio samples, connected to the Smart Walking Stick prototype.
- Metrics: Sensitivity (≥90%), specificity (≤5% false positives), latency (<400ms).

Outdoor Field Simulations

- Purpose: Evaluate real-world performance with variable noise.
- Procedure: Test in parks and streets with simulated distress calls and natural noise (up to 70dB).
- Metrics: Accuracy (≥88%), noise tolerance (<10% false negatives), response time (<3s).

User Validation

- Participant Recruitment: selects diverse users (varying ages, tech experience) for a 1-day workshop.
- Simulation Tasks: includes users calling for help or mimicking falls, logged via the mobile app.
- Feedback Collection: gathers survey responses (target ≥4/5 satisfaction) and verbal input.

Final Validation

- Stress Testing: runs 10-hour tests with high noise (80dB) and temperature swings (10-40°C), targeting >100 hours MTBF and ≥10.5 hours battery life.
- Comparative Analysis: benchmarks against systems like WeWALK, aiming for ≥10% better accuracy and ≥20% faster latency.
- Report Compilation: aggregates data into a presentation for stakeholders, verifying 88% accuracy, <3s response, and 99.8% uptime by June 10, 2026.

4. PROJECT REQUIREMENTS

4.1 Functional requirements

4.1.1 Core Detection

- **FR1:** Capture and process live audio continuously.
- **FR2:** Identify distress phrases with ≥85% accuracy in lab conditions.
- **FR3:** Deliver detection results within 1 second.

4.1.2 Alert Handling

- **FR4:** Trigger vibration/buzzer as immediate confirmation.
- **FR5:** Send SMS to at least one caregiver within 5 seconds of detection.

- **FR6:** Support multiple caregiver contacts.

4.1.3 Communication requirements

- **FR7:** Ensure Bluetooth link between device and Android app.
- **FR8:** Provide offline SMS alerts even without internet.
- **FR9:** Sync detection history to caregiver app via Firebase when connected.

4.1.4 User Interface

- **FR10:** Allow caregivers to add/edit contact details.
- **FR11:** Show detection logs and system status.
- **FR12:** Provide system test mode with sample audio.

4.1.4 Configuration requirement

- **FR13:** Adjustable sensitivity levels (low, medium, high).
- **FR14:** Option to enable/disable haptic feedback.
- **FR15:** Support bilingual recognition (e.g., English and Sinhala).

4.2 Non-functional requirements

4.2.1 Performance requirements

- **NFR1:** Detection latency ≤ 1 second.
- **NFR2:** Accuracy ≥ 80% in noisy conditions.
- **NFR3:** Power-efficient operation for at least 6 hours on one charge.

4.2.2 Reliability requirements

- **NFR4:** SMS delivery success ≥ 95%.
- **NFR5:** Automatic recovery from communication errors within 2 seconds

4.2.3 Usability requirements

- **NFR6:** Setup process ≤ 10 minutes.
- **NFR7:** Caregivers able to operate app with minimal training.

4.2.4 Portability and integration requirements

- **NFR8:** Module weight ≤ 80 g.
- **NFR9:** Operable in -10°C to +50°C.
- **NFR10:** Energy consumption ≤ 500 mA during operation.

4.3 System requirements

4.3.1 Hardware requirements

- **SR1:** Microphone module with ≥50 dB sensitivity.
- **SR2:** Arduino Uno or ESP32 microcontroller.
- **SR3:** HC-05 Bluetooth module.
- **SR4:** Vibration motor & buzzer.
- **SR5:** 2000 mAh rechargeable Li-ion battery.

4.3.2 Software requirements

- **SR6:** Arduino IDE with TensorFlow Lite Micro library.
- **SR7:** Python (Librosa, TensorFlow) for dataset processing.
- **SR8:** Flutter + Firebase SDK for caregiver mobile app.
- **SR9:** Twilio API for SMS notifications.

5. BUDGET AND BUDGET JUSTIFICATION

5.1 Hardware costs

Table 5.1: Hardware costs

Component	Qty	Unit Price (LKR)	Total Cost (LKR)	Justification
Microphone Module (MAX4466)	3	1,200	3,600	Development, testing, spares
Arduino Uno R3	2	3,000	6,000	Core processing units
HC-05 Bluetooth Module	2	1,500	3,000	Redundant connectivity
Hardware Subtotal			**12,600**	

5.2 Software and development costs

Table 5.2**:** Software and development costs

Item	Cost (LKR)	Justification
Twilio API for SMS Alerts	2,700	Used to send automated SMS alerts to multiple caregivers during emergencies.
Cloud Storage (Google Drive Pro)	2,200	Required for maintaining voice datasets and storing experiment results.
Firebase Cloud Messaging	Free	Used for push notifications; free tier is sufficient for research prototype.
TensorFlow Lite / TensorFlow Lite Micro	Free	Open-source libraries for deploying the distress voice detection model.
Arduino IDE & ESP32 Board Support	Free	Development environment for hardware programming.
Flutter SDK	Free	For developing the Android caregiver application.
Software Subtotal	**4,900**	

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